

Not a Playroom, but a Passage: Exploring the Design Space of a Technology-mediated Calm-down Corner

Jiasi Gao
Tsinghua University
Beijing, China
gaojiasi@air.tsinghua.edu.cn

An Chen
Tsinghua University
Beijing, China
chenan@mail.sdu.edu.cn

Zhennan Yi
Tsinghua University
Beijing, China
202011061005@mail.bnu.edu.cn

Xueye Yan
Tsinghua University
Beijing, China
yanxueye@utexas.edu

Jiayi Liu
Tsinghua University
Beijing, China
jiayiliu2019@yeah.net

Guyue Zhou
Tsinghua University
Beijing, China
zhouguyue@air.tsinghua.edu.cn

Jiangtao Gong
Tsinghua University
Beijing, China
gongjiangtao2@gmail.com

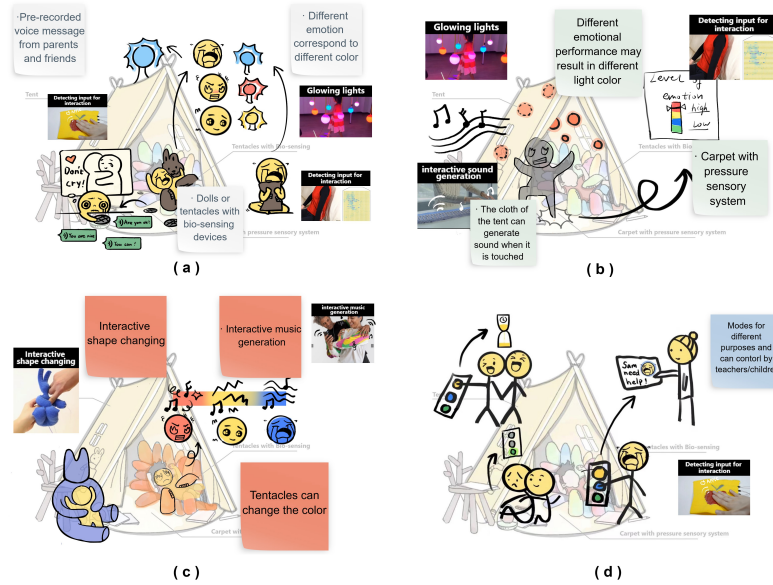


Figure 1: A visual abstract of solutions in smart calm-down corner co-designing workshops with kindergarten educators: a) using color-changeable glowing lights [27] outside the smart calm-down corner for providing intuitive and privacy-protecting visual cue for educators; b) using tactile [16, 17] and biosensor to gather and record indirect emotion information of children for real-time and long-term use; c) providing restrained feedback [1] to regulate children's emotions; d) offering mode-switch options to adjust the interaction mode for different usage.

ABSTRACT

Preschool educators play an important role in supporting children's emotion regulation capacities in ways that promote their socio-emotional and behavioral health. However, tracking children's emotional changes and giving timely intervention to every child in the class simultaneously is challenging for preschool educators. In this paper, we use the "calm-down corner," which is widely used in classrooms with spaces where children can regain emotional control, as a probe to explore the design space of technology-supported

preschool educators dealing with children's emotional problems through one semi-structured interview with ten preschool educators and one co-design session with another 12 preschool educators. We summarized current challenges and the design considerations from preschool educators' perspective, which could offer valuable insights for technology-enabled systems to facilitate the emotion regulation process for educators in the future.

CCS CONCEPTS

• **Human-centered computing** → **Participatory design**;

KEYWORDS

emotion regulation, educators, qualitative research, preschooler

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1 INTRODUCTION

Emotion regulation (ER), as a crucial aspect of emotional competence, is essential for child development in mental health, language, academic performance, and social behaviors [7–9, 24, 25]. Preschools and kindergartens are the settings where children spend most of their time outside of their families. As in their early childhood, preschoolers learn and develop ER skills through different situations. Various relationships and activities in the preschool scenario allow them to fully encounter different conflicts and learn to resolve the resulting emotional issues. Educators play an essential role in these possible educational opportunities. Although educators contribute greatly to the development of children's lifelong well-being, it is still challenging for them to understand each child's emotional experiences and provide adequate support and timely response to the child's emotional needs [2, 10].

In recent years, increasing numbers of studies aimed at gaining a deeper understanding of how educators foster children's emotion regulation skills. These findings highlight key aspects such as the quality of the educator-student relationship [6, 19, 21], the educator's level of emotional intelligence [26], and education towards emotion regulation [22]. From the human-computer interaction (HCI) perspective, previous interventions for child emotion regulation mainly focused on emotion recognition [13, 20], emotion expression and communication [4, 5, 27], and emotion regulation training [3, 15]. What is evident in all these works, however, is that understanding children's emotions is not a straightforward task. It is also essential and challenging for preschool educators to help children with their ER process [18]. This emotional communication gap identifies the area where technologies might support emotion regulation in classrooms.

The "calm-down corner" (or "peace corner") is a physical location that can be used to help students regain emotional peace when experiencing an emotional breakdown, which is already widely used and closely related to ER and ER education in preschool and kindergarten [14, 18]. Many different tools can be introduced into

a calm-down corner to help students calm down and get back to the classroom mindset, such as having comfortable options for seating, sensory calming tools, and activities [18]. Previous studies have shown that the calm-down corner can help children learn to master the skills of ER and also help educators reduce the stress of teaching ER skills in class [11]. To this end, in this paper, we choose the technology-augmented calm-down corner as a design probe to explore possible technologies to facilitate educators' foster children's emotion regulation in a preschool context. Our study is aimed to investigate two important research questions:

- QR1: What challenges do educators have when they use the calm-down corner to help children with emotional problems?
- QR2: How to use smart interactive technologies to help educators with these challenges?

To address these questions, we conducted a mixed-method study with 22 preschool educators using semi-structured interviews and participatory design. We first interviewed 10 educators to understand the current practices and challenges of the calm-down corner. The needs analysis created the design space for a technology-mediated calm-down corner. Then, we conduct co-design workshops with another 12 educators. The educators and our researchers in the co-design workshop could propose ideas for smart calm-down corners and discuss them with each other from their own experiences. Based on these results, we extracted broader design implications for interactive systems so that the smart calm-down corners can facilitate educators' work on children's emotion regulation.

The contributions of this paper include: 1) a deep understanding of the current practices and challenges of preschool and kindergarten educators who use the calm-down corner in children's emotion regulation processes. 2) Possible technology solutions for smart calm-down corners that support better ER processes and education for children. 3) Summarized design considerations from preschool educators' perspective to provide insights for future emotion regulation technology in preschool.

2 METHODS

Based on our research objectives, we have adopted a combined approach of semi-structured interviews and participatory design. The use of semi-structured interviews in the first phase allows researchers to engage in in-depth conversations with practitioners to better understand their viewpoints, experiences, and insights, which also serves as valuable preparation for exploring future emotion corner design spaces in the second phase. In the second phase of participatory design, researchers, product designers, and end-users will come together in workshops to propose feasible design solutions for a smart calm-down corner. As the result, we had 22 kindergarten educators participate in studies (see in Appendix, Table 1), 10 of whom (P1-P10) participated in the semi-structured interviews (Study 1), and 12 (P11-P22) participated in the participatory design sessions (Study 2).

In Study 1, all participants were from one public preschool located in Beijing, China. Throughout their teaching periods, calm-down corners were always available in their classrooms. In Study 2, participants came from a diverse range of kindergartens including Public, Private, Bilingual, and Montessori schools across different

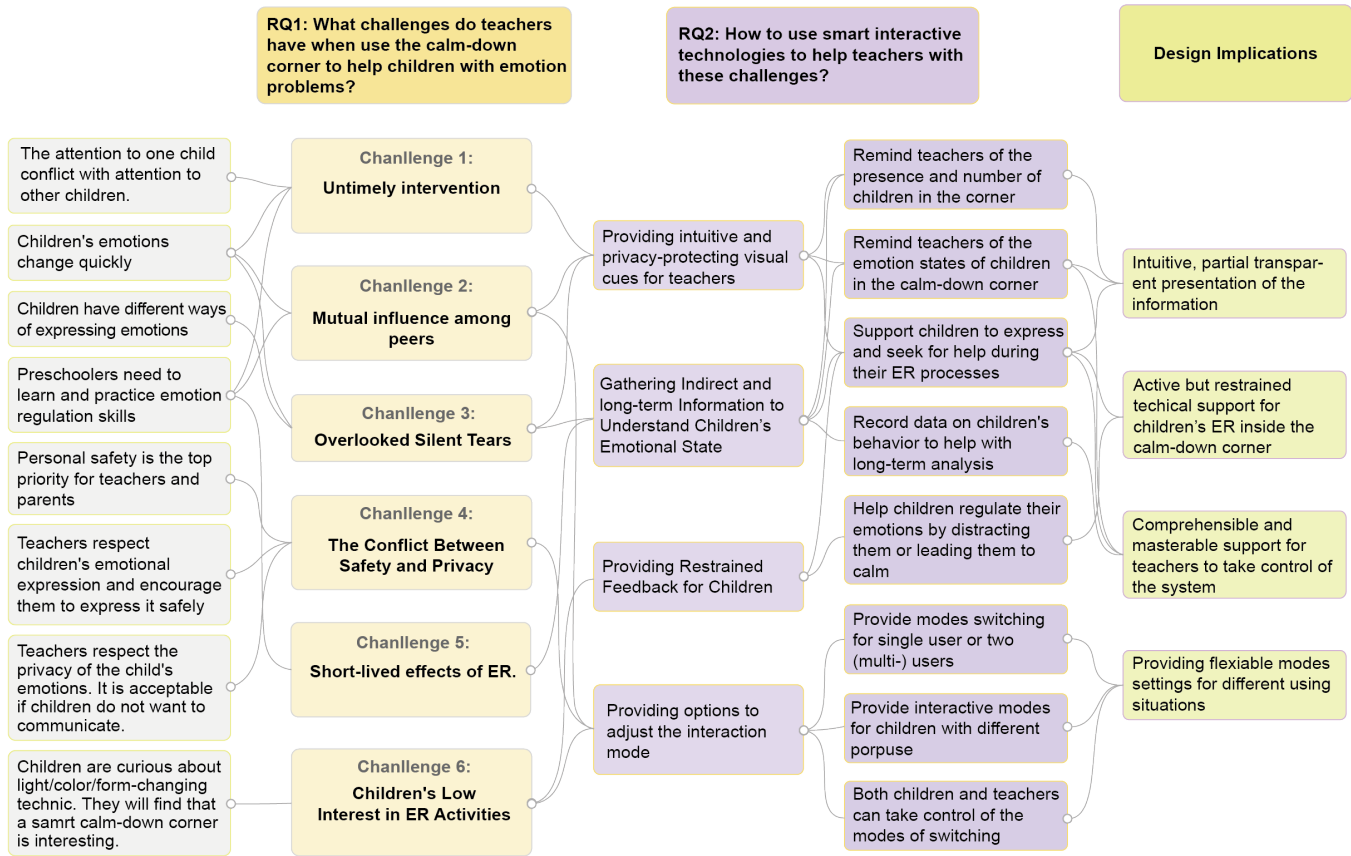


Figure 2: Thematic analysis map of findings.

regions in China. Only some of them mentioned having calm-down corners established within their classroom environments.

2.1 Procedure

Semi-structured Interviews Sessions Sample interview questions included "How does the calm-down corner look like?", "How was it designed", "How would children with different characters behave when they have negative emotions, do they prefer to go to the calm-down corner?" and "What are the pros and cons of the corners?". Educators were also asked to demonstrate pictures of the calm-down corner in their classes (see in Appendix, Fig.4). We dug for detailed questions about the role of the calm-down corner in the process of soothing children and ER skills education, as well as the difficulties and challenges educators encountered when using the calm-down corners.

Participatory Design Sessions We followed up our interviews and focus groups with 4 participatory design sessions to co-design solutions with participants. Participatory design sessions were held with educators sizes of 2 to 4. Four researchers with backgrounds in design, education, and psychology were invited to participate in each participatory design session.

The workshops were conducted remotely through the online video conferencing tool Tencent Meeting, which meets data protection legislation and university ethics standards. We used an online

Whiteboard application as design space. This tool works as a collaborative canvas and allows participants to easily see, write, draw, and drag existing design elements on the shared space. The workshops were audio and video recorded according to participants' consent. Two researchers as facilitators were available throughout the virtual workshop to help record their ideas on the shared canvas.

At the beginning of the workshop, participants were given a brief overview of the project and shown a preliminary prototype of a smart calm-down corner, developed based on insights gathered from Study 1. This prototype depicted a scenario of a child utilizing the calm-down corner to manage his emotions. Design spaces for the interactive system, such as the tent, tentacle-patterned carpet, and surrounding area, were introduced. Additionally, various design elements were provided based on insights from Study 1 and related research, including methods for detecting physiological signals, visual and auditory solutions for emotional regulation, and interactive technology in soft textures. Then, following the introduction, participants were prompted to reflect on their experiences handling children's negative emotions and generate ideas for the smart calm-down corner's functionality. Brainstorming sessions led by researchers covered topics such as information educators should observe about children, comforting feedback, privacy considerations, and parental involvement, etc. Throughout the workshops, participants collaborated to design how an interactive system would

collect and respond to children's emotions. Finally, participants explored solutions to facilitate children's return to daily activities and enhance educators' information acquisition efficiency. Each design workshop continued for around 2-3 hours, depending on the number of participants.

2.2 Analysis

All study data were screen-recorded via Tencent Meeting and auto-converted into text, then double-checked by the authors. Participants' Whiteboard ideas were documented in a spreadsheet. Themes were identified using inductive open coding and thematic analysis, focusing on semantic data interpretation. Initial coding was done by two authors, with research team members discussing the conceptualization of the topics and unifying the codebook for multiple rounds of coding.

3 FINDINGS

Through interviews, focus groups, and participatory design sessions, we identified 6 challenges educators experienced as they worked with calm-down corners. These challenges were seen as design considerations that needed specific solutions. Fig. 2 shows the thematic analysis map of findings.

3.1 Study 1: Challenges of Calm-down Corners in Classrooms

We found that for educators who have an emotion corner in the classroom, the frequency of using the calm corner is quite high in their daily routine. Educators adopt different emotion regulation strategies and use the calm corner differently based on the children's characteristics in emotional expressions, such as introverted and extroverted types [12]. Fig.3 obtained through qualitative analysis from discussions with multiple participants, visualizes how children with different emotional expression types use the calm corner. When educators perceived children's negative emotions, they used the "See the emotion - Address the emotion - Solve the problem - Strengthen emotion regulation skills" approach. In this section, we introduce 6 themes to help frame the challenges of the emotion regulation process by using the calm-down corner in the kindergarten context.

Untimely Intervention. Although most educators felt that the calm-down corner was effective in helping them relieve stress by pacifying the children, they still acknowledge that educators have limited attention and may resort to a "delayed treatment" strategy. While sometimes "let the child cry for a while, and their emotions will improve" (P7), sometimes it can lead to "the child's emotions getting out of control" (P10), potentially leading to more complex emotional issues. Preschool children experience rapid emotional changes, necessitating educators to identify opportune moments for ER interventions that are suitable for both themselves and the children. Most participants report a desire to spend more time seizing each opportunity for emotional regulation with the children, educators in practice can only selectively intervene in emotional events they deem urgent or significant due to practical constraints. Consequently, they express hope for the development of the smart calm-down corner to provide timely support for their ER interventions.

Overlooked Silent Tears. Preschool-age children are in the learning phase of ER, which also means they may fail in their ER attempts. Also, with different personalities and expression development levels, some are not good at verbally expressing their emotional problems. For example, "Children would sit by him/herself silently in the calm-down corner holding stuffed animals and looking at the educator from time to time." (P9) Therefore, educators may not be able to promptly detect these children's negative emotions. Therefore, educators expected that a smart calm-down corner could help them with this overlooked situation so that can serve more to help educators notice those easily overlooked children.

Mutual Influence among Peers. Peer socialization is a double-edged sword in the ER process. Some educators reported peers' comfort was sometimes more effective than educators' involvement in children's ER processes. Sometimes, "other kids may give him/her a sticker, a treasure, or tell him/her that they would go together to a magic castle, sharing secrets belonging to their own. It won't take long to divert the child's attention and change his or her mood." (P5) However, sometimes peer social interactions can trigger a chain of emotional events:

"Two girls in my class are very good friends. One day, one of them accidentally made the other one cry. The girl was panicked and kept saying, 'I'm sorry! Do not cry, please!' However, the crying girl did not stop. Then the girl who consoled said, 'Sorry, I already said sorry to you. How are you still crying?' Then she also began to cry, and both kept crying until the educator's intervention." (P7)

Information and strength from peers can naturally connect children to social activities and thus help them return to everyday life. Thus, most interviewees agreed that incorporating children's social attributes into the ER process is positive if a smart calm-down corner could involve rules or technical means to alert educators promptly to enter the ER process.

The Conflict Between Safety and Privacy. The personal safety of preschool children is the top priority for educators and parents. Educators stated they must ensure that all children are in sight according to the kindergarten and parents' requirements. However, to provide a relatively independent and private space, calm-down corners are almost semi-open settings, which makes it more difficult for educators to ensure the safety of children. Therefore, the educator needs to "keep observing" (P2) and constantly evaluate the state of the children in the calm-down corner through visual and auditory means to ensure that they are safe. Hence, ensuring that children feel their privacy is protected while also ensuring that educators are constantly aware of the emotional state of children within the calm-down corner is a challenge.

Children's Low Interest in ER Activities. The role of the emotion corner is to "help children adjust their emotions" (P10) and "return to their daily activities" (P20). Educators appreciate the intervention concepts and potential approaches within the calm-down corner. However, some of them stated that "children's attention is more easily drawn to colorful and moving things" (P3). Hence, they were concerned that children would use the smart calm-down corner as their new "activity zone," potentially delaying their ER needs. Thus it is important to design the interface feedback to regulate their emotions and encourage them to return to the class when appropriate.

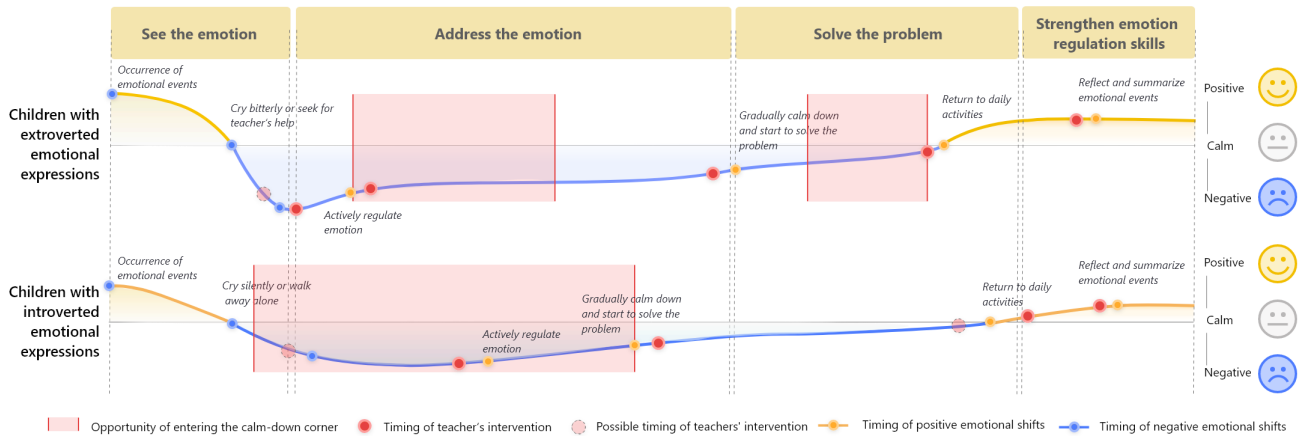


Figure 3: The processes of using the calm-down corner for children with extroverted emotional expression and with introverted emotional expression.

Short-lived Effects of ER. As previously mentioned, preschool-age children’s emotions change relatively quickly, and fewer negative emotions persist for long periods. During emotional events, many children are easily distracted and tend to forget their ER experiences. Participants believe that although educators instruct ER skills in the classroom, not all children exhibit significant changes within a semester or a year. Additionally, some participants view each ER experience as a critical opportunity for children to learn ER skills. Therefore, they sometimes “review” past ER experiences when the child’s emotions are stable (P8), to reinforce their ER skills. However, according to participants’ descriptions, the multitude of emotional events occurring daily in kindergartens makes it challenging for educators to remember all children’s ER experiences based solely on memory. Moreover, when discussing a child’s development and coping strategies with parents, they lack visual data and developmental trajectories, which hinders effective communication and strategy formulation.

3.2 Study 2: Designing the Interactive System for Smart Calm-down Corner

By examining educators’ points of view in study 1, we highlighted the need for the smart calm-down corner to be an effective “passage” to: 1) empower children with negative emotions to recover and return to daily activities; 2) provide educators with compelling information on the emotional states of children; 3) provide educators with an emotional education environment/foundation that is easily adaptable to the diverse needs of students; 4) provide children, educators, and parents with comprehensive long-term tracking information about the child in the clam-down corner.

We address the above needs for technology solutions with participating educators in study 2. Overall, the participant educators believe that adding technology into the calm-down corner can benefit children’s ER process, and they discussed possible solutions and concerns from their experience.

In study 2, participants proposed smart emotion corner concepts as shown in Fig. 1. Educators appreciated lighting and sound design, like Fig. 1 (b), using them to “alert” (P9) about children’s presence. As shown in Fig. 1 (a), participants also suggested “matching the color of the lights with the children’s emotions” (P5). Some proposed recording tracks in the calm-down corner to monitor children’s emotions and behaviors for better ER teaching. Many favored soft dolls or beanbags as shown in Fig. 1 (c) to help children “vent” (P4) and “distract” (P5). An hourglass was suggested by a few to aid in “calming down” (P21). Additionally, the mode-switching buttons were designed for changing play modes, see Fig. 1 (d). Besides the smart corner concept design, participants also shared their design considerations. We summarized these design considerations into four solution themes.

Providing Restrained Feedback for Children. In the workshop, educators proposed three possible supportive ways: interactive environmental factors, proactive tactile feedback, and targeted message support.

Interactive environmental factors: Most participants stated that children could already acknowledge that different colors could represent different emotions and tell emotions the music conveys. Specifically, some educators suggested that utilizing light’s color and music’s rhythm changes would benefit children’s ER, see Fig. 1 (c). “Once the angry boy/girl steps in, the music rhythm will play based on his/her heartbeat rate. Then the music slowly adjusts to a calm rhythm, implicitly guiding the child to a peaceful state.” (P5)

Proactive tactile feedback: Besides, most educators agreed that tactile feedback is essential for the smart calm-down corner. They emphasized the high frequency of physical interaction with children during the daily emotion regulation process. Although most educators felt that providing hugs for any child with bad emotions is necessary, this feature was seen as more desirable “for children who are relatively introverted” (P13), because if a big doll can hug him without asking why, it will be very calming to him.

Targeted message support: Given the power of peers and the different causes of emotional issues, some educators suggested

providing restrained "outsider" interventions in the calm-down corner is valuable, as shown in Fig. 1 (a). However, a few educators argued that while the visual way of showing photos or videos could cause a more profound sorrow of missing, an auditory approach can contribute to supporting children back to their daily activities.

Overall, these suggest the need for a smart calm-down corner with a restrained design that can respond to the stakeholders' needs and situations' constraints.

Providing Intuitive and Privacy-protecting Visual Cues for Educators. Educators must monitor every child's safety and give those who need it as soon as possible despite their short attention spans. Four types of information are mentioned by educators: the presence or absence of children in the calm-down corner; the child's safety status; the child's real-time emotional status; and the child's active or passive message for help.

However, a calm-down corner is generally located relatively far from the main activity area in the classroom. Thus, how to quickly gather that information at a distance is mentioned by 4 out of 10 educators mentioned. The educators prefer visual cues to audio cues because there are often various musical sounds in the kindergarten to help children make activity switches. Thus, educators agreed that a comprehensive visual representation of real-time children's states could support them in knowing the children's overall states in a calm-down corner. For Children, our participating educators believed that intuitive visual cues like color-changeable light can create a medium for triggering emotional communication between educators and children while allowing children to feel that they are noticed and cared for by educators. Most educators feel that discussing and understanding children's emotions in a kindergarten context is much easier when there is a medium that serves as an entry point for discussing emotions.

Providing Options to Adjust the Interaction Modes. The necessity for peer socialization and privacy protection, as well as the positioning of the emotion corner, indicates that calm-down corners need multi-modes. As shown in Fig. 1 (d), participants summarized four modes in the co-design session by combining the perspectives of the number of children, the child's privacy, and the purpose of use.

Self mode is for one child who is willing to regulate his/her emotions alone. When entering the emotion corner, the system implicitly activates to collect the child's primary and interactive data. The system would inform educators when it detects something unusual or if the child calls for help in any way.

Peer mode is when other children join in with a particular child's emotion regulation process. When two or more children are in the calm-down corner, the central part of the regulation is their interaction. At the same time, the system mode should turn from a "detecting and comforting" mode to an "observing and providing soothing environment" mode. educators suggested that the system should "tell them the number of children and keep the channel open for them to ask for help at any time".

Fun mode emerged from the primary usage of calm-down corners. Participants sometimes found that children go to the calm-down corner just for fun, which would be more frequent if "lights and music could change and even toys could hug" in their belief. Thus, P13 and P22 proposed utilizing a digital hourglass on the

tent's wall to meet their playing needs while ensuring that the space is not occupied all the time.

Privacy mode is designed for children who do not want their physiological data to be collected in any way or want complete quietness. The educators agreed that all the child's rights should be respected while ensuring safety. Although some claimed that the privacy mode of a smart calm-down corner is just like their current, no-tech one, participants felt they needed to be extra attentive to these children.

The participants also defined the users who could switch modes. They argued that in terms of timeliness, the system should prioritize determining which mode needs to be activated, secondly, the child holds the right to switch, and lastly, the educator.

Gathering Indirect and Long-term Information to Understand Children's Emotional States. To better intervene in children's ER process and educate their ER skills, the participant educator suggested using some indirect method (different from a surveillance camera) to gather and record children's emotion, such as tactile and physiological sensors, see Fig. 1 (b)(c). In the current kindergarten context, feedback on the effectiveness of education can only come from a small number of parents or long-term observations by educators. By recording behavioral data, educators could see changes in children's behavior over a shorter period and thus make more timely adjustments to their emotional education strategies for each child. Meanwhile, some educators said the records should be presented visually to help them review the ER growth with parents and children.

4 DISCUSSION

In previous studies, HCI researchers have focused on exploring the interactions between children and parents, while psychologists and educators have also explored physiological basis and association mechanisms among different groups in the emotion regulation research area. Our work presents the first example of research looking specifically at how to support educators in interacting with the smart calm-down corner.

The findings across two studies reveal challenges and design considerations for emotion regulation, especially for preschool educators. They suggest an interactive system using multi-sensory channels for children with varying personalities and development levels, aligning with the emotional regulation traits of this age group [23]. Children's fluctuating emotions require guidance through tactile, auditory, and visual stimuli for attention shifting. Yet, there's concern over children's overindulgence in such interactions, leading to a design tension between diverse information and the need to integrate back into daily life/class. These insights aid in developing future technology-mediated emotion regulation education for young children.

5 LIMITATIONS

Educators in Study 1 were from the same kindergarten at the university. Therefore, our study sample may be biased toward the kindergarten educator population in China. Educators in Study 2 were recruited via social media platforms. Educators who signed up for the interview expressed their strong interest in child emotion regulation, while some educators canceled the interview because

they knew little about the topic. A selection bias occurred because educators who showed no interest in ER or were unaware of the importance of ER might be excluded. Further, one or two educators from certain kindergarten types and certain cities participated in the study, which may be hard to generalize their needs to all educators' needs in China. In addition, the results presented may include gender bias due to the absence of male participants in our studies.

6 CONCLUSION

Emotional regulation education is crucial for young children and is also an important task for preschool and kindergarten educators. A technology-mediated calm-down corner creates additional interactions while in the expectation of addressing the previous needs of educators. By conducting a series of semi-structured interviews and design workshops, we explored for the first time the users' and stakeholders' needs for smart calm-down corners in kindergarten. It is important to note that the calm-down corner provides not only a space to solve emotional issues but also a passage to better prepare children for classroom activities after emotional events occur.

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A PRE-STUDY DETAILS

A.1 The calm-down corner pictures from our study participants

B EXPERIMENT DETAILS

B.1 Demographic information of participant educator



Figure 4: The calm-down corner pictures from our study participants.

Table 1: Participant demographic and background information

ID	Gender	Age	Years of Teaching	Class age level	Types of Kindergarten	City
P1	Female	43	12 years	The middle class	Public	Beijing
P2	Female	38	16 years	The middle class	Public	Beijing
P3	Female	32	10 years	The senior class	Public	Beijing
P4	Female	34	12 years	The junior class	Public	Beijing
P5	Female	34	7 years	The junior class	Public	Beijing
P6	Female	43	21 years	The senior class	Public	Beijing
P7	Female	30	9 years	The middle class	Public	Beijing
P8	Female	28	6 years	The senior class	Public	Beijing
P9	Female	27	6 years	The junior class	Public	Beijing
P10	Female	29	7 years	The middle class	Public	Beijing
P11	Female	34	10 years	The middle class	Public	Taizhou
P12	Female	31	8 years	The top class	Public	Taizhou
P13	Female	35	7 years	The top class	Private	Hangzhou
P14	Female	23	2 years	The middle class	Public	Hangzhou
P15	Female	40	20 years	The top class	Public	Yanzhou
P16	Female	48	30 years	The bottom & middle & top class	Public	Beijing
P17	Female	35	11 years	The bottom class	Public	Beijing
P18	Female	27	4 years	The middle class	Bilingual & Private	Weifang
P19	Female	35	14 years	The top class	Public	Shanghai
P20	Female	41	18 years	The bottom & middle class	Bilingual & Private	Beijing
P21	Female	27	5 years	The top class	Private	Weifang
P22	Female	27	5 years	The bottom class	Montessori	Qingdao